



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-2024
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL APPLICATIONS
[SET-A]

FULL MARKS: 100
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of six printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

Question 1

Choose the correct answers to the questions from the given options.
(Do not copy questions, write the correct answers only.)

[20]

- i. Extensive planting of trees to increase cover is called
 - a) Agroforestry
 - b) Social forestry
 - c) Deforestation
 - d) Afforestation
- ii. It is difficult to recycle plastics because
 - a) it is hard
 - b) it comes in different sizes
 - c) it is adhesive in nature
 - d) it contains different types of polymer resins
- iii. Recycled paper is banned for use in food containers as
 - a) it creates lot of spaces
 - b) it creates contamination
 - c) paper can be used only once
 - d) paper is very thick and cannot cover the food containers
- iv. Water scarcity is not due to
 - a) rapid growth of population
 - b) uneven distribution of water resources
 - c) construction of dams and reserves
 - d) increase in demand
- v. The Bhakra Nangal Project is the largest multi-purpose power project in India. Choose the option that is not in favour of this project.
 - a) It brings water to those areas which suffer from water scarcity.
 - b) It regulates water flow to control flood.

- c) It leads to large scaledisplacements and loss of livelihood.
- d) It generates electricity for our industries and homes.
- vi. A process of deposition of soil particles in the waterways and dams is
 - a) leaching
 - b) siltation
 - c) filtration
 - d) erosion
- vii. Assertion: Recycling is the best solution for the treatment of E-wastes.
Reason: E-wastes are often buried in landfills or incinerated.
Choose the correct option
 - a) A is true and R is the correct reason for A.
 - b) A is false and R is not the correct reason for A.
 - c) A is false but R is true.
 - d) Both A and R are true.
- viii. The system that processes waste before discharging it to the environment is
 - a) Screening
 - b) EOP
 - c) Open dumping
 - d) Landfill
- ix. The area of the TTZ around Taj Mahal is
 - a) 10,400 sq.km
 - b) 100,40 sq.km
 - c) 100,400 sq.km
 - d) 1400 sq.km
- x. The vehicle which uses two or more distinct power sources
 - a) Electric vehicle
 - b) Cycle rickshaw
 - c) Solar car
 - d) Hybrid vehicle
- xi. An alternative to fuel wood in paper industry is
 - a) biogas
 - b) log
 - c) bagasse
 - d) metals
- xii. “Catch water where it falls” is the idea behind
 - a) recycling of water
 - b) watershed
 - c) rainwater harvesting
 - d) catchment of water
- xiii. Study the picture given below and answer the question that follows



Identify the person in the picture who is rightly called ‘Water man of India’.

- a) Sundarlal Bahuguna
 - b) Rajendra Singh
 - c) Vinoba Bhave
 - d) Anna Hazare
- xiv. Check dams are
- a) one of the soil conservation strategies
 - b) one of the soil erosion methods
 - c) meant to generate electricity
 - d) meant to reduce wind speed
- xv. Activated sludge is _____ for treating effluents.
- a) floc of bacteria, fungi protozoa
 - b) sediment of soil particles
 - c) wet mud
 - d) group of chemical compounds
- xvi. A forestry where both the forest official and local people are involved
- a) Social forestry
 - b) Community forestry
 - c) Agroforestry
 - d) JFM
- xvii. Reasons for necessity of timber alternatives is
- a) Deforestation
 - b) Earthquake
 - c) water pollution
 - d) soil pollution
- xviii. Remote sensing satellites are placed
- a) on the ground
 - b) in the outer space
 - c) in the ocean
 - d) under the ground
- xix. The industrialised world has threatened our
- a) population
 - b) biodiversity
 - c) pollution
 - d) recreation activity
- xx. A greenhouse gas produced in landfill is
- a) Carbon dioxide
 - b) Methane
 - c) Water vapour
 - d) Nitrogen

Question 2

- a) How does soil erosion affect navigation? [2]
- b) There are lots of social, environmental and economic reasons to use recycled water. Mention some ways to use recycled water. [2]
- c) Euro norms are vehicle emission standards for pollution from the use of new land surface vehicles and sold in the European Union. They are stringent standards but with some limitations. State three such imitations of Euro norms. [3]

d) Study the picture given below and answer the question that follows

[3]



What does the marked area indicate? Suggest ways to protect the heritagesites within the marked zone.

Question 3

- a) Much of the world is constantly trying to find sustainable and environment friendly energy solutions to move us and power our everyday lives. However, there are still challenges to overcome. In this context, give two reasons why alternative fuels are not popular in India. [2]
- b) What is mulching? How is it helpful? [2]
- c) Give two reasons why construction wastes are harmful. Give two examples of the same. [3]
- d) Where is Curitiba located? State any two traffic management policies which we can learn from there. [3]

SECTION B (60 MARKS)

(Attempt any four questions from this section)

Question 4

- a) Why are remote sensing satellites considered good tools for gathering environmental information? [5]
- b) India is heavily confronted by the lack of natural resources and is unable to provide its citizens with enough resources. Natural resource depletion has grave environmental consequences. Discuss. [5]
- c) 90 acres of forest are destroyed everyday across the world. What are the possible causes for such destruction? [5]

Question 5

- a) Many major cities across the globe are at risk of water crisis, and water stress is projected to increase in most countries in the coming decades, threatening regional stability and raising the possibility of forced migrations. Suggest five ways to mitigate the problem. [5]
- b) How can economic policy measures help to control vehicular pollution? [5]
- c) Which city is known as the 'City of Dawn' and why? [5]

Question 6

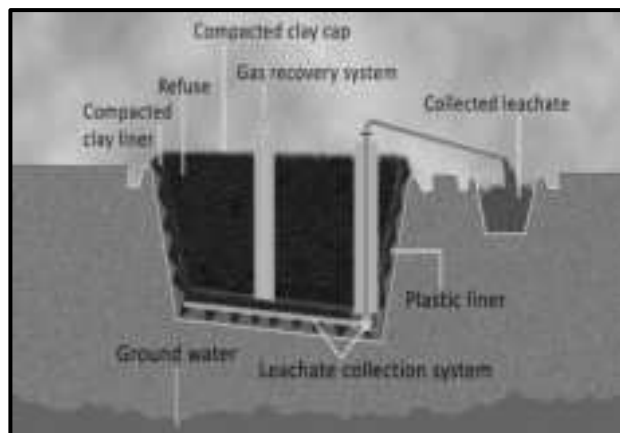
- a) Community participation brings many lasting benefits to people. Discuss. [5]
- b) The East Kolkata Wetlands, are a complex of natural and human-made wetlands lying east of the city of Calcutta. They cover 125 sq km and is ecologically and economically very important ecosystem. State five reasons why wetlands should be protected. [5]
- c) How would you spread awareness about the environment in your neighborhood? [5]

Question 7

- a) Most of us claim to have read up on waste management processes. However, it should come as no surprise that despite our best efforts, the damage has continued, only at a different rate. We have come a long way in efficiently dealing with the waste, yet several key problems continue to merge and injure the planet at large. Discuss the problems of waste disposal in India. [5]
- b) Name the pollution control device which separates pollutants by using the following principles: [5]
 - Electrical charge
 - Centrifugal force
 - High temperature combustion
 - Basic sprayWhat is the method known as which uses all these devices to control pollution?
- c) A large variety of alternative fuels and additives have evolved as a choice to be used in India. Discuss. [5]

Question 8

- a) Study the picture given below and answer the following questions



Identify the above method of waste disposal. How is it a better option than open dumping? [5]

- b) For decades, we have failed to design our buildings with health as the primary focus, thus leading to severe indoor air pollution. How can you minimize indoor air pollution? [5]
- c) Through an ordinance titled Tamil Nadu Municipal Laws Ordinance, 2003, dated July 19, 2003, the government of Tamil Nadu has made rainwater harvesting mandatory for all the buildings, both public and private, in the state. Explain the method of modern rainwater harvesting system followed. [5]

Question 9

- a) The population of our country is very large whereas, the land fit for cultivation is limited. If you are a farmer what methods would you adopt to improve your crop yield? [5]
- b) Hazardous waste transportation is given utmost importance. Hazardous waste has, in the past, been intentionally spilled or abandoned at random locations in a practice known as 'midnight dumping'. This practice has been greatly curtailed by the enactment of laws.
What are hazardous wastes? Name the Convention which addressed the issue of hazardous wastes. What were its objectives? [5]
- c) As the transmission of novel corona virus increased rapidly, the whole world adopted the lockdown policy. It was found that the pollution level reduced during the lockdown. Comment. [5]